



Cheat Sheet: Arduino Programming Basics – Final Exam MIAE 215

Course: [Mechanical, Industrial & Aerospace engineering \(MIAE 215\)](#) 378 documents University: [Concordia University](#)  Zakaria Chaibat 2025/2026

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This document outlines the structure and functionality of Arduino programming, covering essential concepts such as setup and [...more](#)

Arduino

Program structure:
setup(): runs once
loop(): runs repeatedly
Serial:
Serial.begin(115200);
Serial.print(x);
Serial.println(x);
Time:
delay(t);
millis(); returns ms since start
Non-blocking timing: if(millis() - last >= T) last = millis();
Digital I/O:
pinMode(p, OUTPUT);
pinMode(p, INPUT_PULLUP); // HIGH = not pressed, LOW = pressed
digitalWrite(p, HIGH/LOW);
digitalRead(p); // returns HIGH or LOW
Analog input:
analogRead(A0); // returns 0–1023 for 0–5V
Example: v = analogRead(A0);
Servo:
#include <Servo.h>
Servo s;
s.attach(pin);
s.write(angle); // 0–180°
Common patterns:
// LED control
if(v > 600) digitalWrite(13, HIGH); else digitalWrite(13, LOW);
// Button (INPUT_PULLUP)
if(digitalRead(7) == LOW) { /* pressed */ }
// Map sensor to servo
v = analogRead(A0);
angle = v / 6;
s.write(angle);
// Angle limits
if(angle > 180) angle = 180;
if(angle < 0) angle = 0;
Examples:
// Print sensor
v = analogRead(A0); Serial.println(v);
// Button controls LED
pinMode(7, INPUT_PULLUP);
pinMode(13, OUTPUT);
if(digitalRead(7) == LOW)
digitalWrite(13, HIGH); else
digitalWrite(13, LOW);
// Servo every 500 ms
if(millis() - last >= 500) {
last = millis();
s.write(angle);
}
EXAMPLE OF QUESTION :
Uses a pushbutton on **pin 7** with INPUT_PULLUP.
Turns an LED on **pin 13** ON when the button is pressed (LOW), and OFF otherwise.
Reads a potentiometer on **A0**.
Every **200 ms**, print the analog value to Serial at 115200 baud.
No delay() allowed → must use

```
#include <Arduino.h>

unsigned long lastTime = 0;

void setup() {
  Serial.begin(115200);

  pinMode(7, INPUT_PULLUP); // button
  pinMode(13, OUTPUT);      // LED
}

void loop() {
  int button = digitalRead(7);

  // LED control
  if(button == LOW)
    digitalWrite(13, HIGH); // pressed
  else
    digitalWrite(13, LOW);  // not pressed

  // timing using millis()
  unsigned long t = millis();
  if(t - lastTime >= 200) {
    lastTime = t;

    int val = analogRead(A0); // 0-1023
    Serial.println(val);
  }
}

Example with servo
#include <Servo.h>
Servo servo1;
unsigned long lastT = 0;
void setup(){
  servo1.attach(9);
}
void loop(){
  unsigned long t = millis();
  if(t - lastT >= 150){
    lastT = t;
    int v = analogRead(A0); // 0..1023
    int angle = v * 180 / 1023; // map to 0..180
    servo1.write(angle);
  }
}

Arrays
Declaration:
int a[100];
double x[50];
char name[20];
Initialization:
int a[5] = {1,2,3,4,5};
double x[] = {1.1, 2.2, 3.3};
Indexing:
a[0] = first element
a[i] = i-th element
Valid indices: 0 to n-1
Common array loop pattern:
for(int i=0;i<n;i++) { ... }
Reading into array:
int n;
int a[n];
```

Modifying all elements (example: multiply by 3):
for(int i=0;i<n;i++) a[i] = a[i] * 3;
Finding maximum:
max = a[0];
for(int i=1;i<n;i++)
if(a[i] > max) max = a[i];
Sum and average:
sum = 0;
for(int i=0;i<n;i++) sum += a[i];
avg = sum / n;
Algorithm:
for(i=0;i<n-1;i++)
for(j=0;j<n-1-i;j++)
if(a[j] > a[j+1]) swap(a[j], a[j+1]);
Swap function:
tmp = a[j];
a[j] = a[j+1];
a[j+1] = tmp;
Question : read nb from a file into an array and bubble sort
n=0; while(fin >> a[n]) n++;
for(i=0;i<n-1;i++)
for(j=0;j<n-1-i;j++)
if(a[j] > a[j+1]) swap(a[j], a[j+1]);
cout << a[0] << " " << a[n-1];
Pointers :
Declaration:
int *p;
double *q;
Assign pointer to a variable:
p = &x; // p holds the address of x
*q means "the value stored at the address q points to"
Important operators:
&x → address of x
*p → value pointed to by p
p = &x → pointer p now refers to x
*p = 10 → changes the variable p points to
Pointers and arrays:
If a is an array, then:
a == &a[0]
p = a; // p points to element 0
p+i points to a[i]
*(p+i) == a[i]
p[i] == a[i]
Common pointer arithmetic:
p++ moves to next element in the array
p+i moves i elements forward
Passing pointers to functions:
void f(int *p) { *p = 10; }
f(&x); // x becomes 10
Passing the address allows the function to modify the original variable.
Passing arrays to functions:
void f(int a[], int n) { ... }
Same as using a pointer: a is treated as int * inside the function.
Dynamic memory (if needed):
p = new int[n];
delete [] p;
Typical pointer mistakes to avoid:
Never dereference an uninitialized pointer.
→ go out of array bounds (p+i must d)

 Mock exam  Summary  Quiz

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```
int p = (a+i)
p[i] == *(p+i)
p = a; // p now points to element 0
*(p+2) == a[2]
Example 1:
int x = 5;
int *p = &x;
*p = 7;
// x becomes 7
Example 2 (array pointer):
double a[4] = {1.1,2.2,3.3,4.4};
double *p = a;
p[2] = 9.9;
// a[2] becomes 9.9
Example 3 (pointer + function):
void inc(int *p){ *p = *p + 1; }
int x = 4;
inc(&x);
// x = 5
Example 4 (moving pointer across array):
int a[5] = {3,6,9,12,15};
int *p = a;
p = p + 3;
// p points to 12
*p = 99;
// a[3] = 99
Function prototype:
return_type name(parameters);
Example:
int sum(int a, int b);
Function definition:
return_type name(parameters) {
// code
return value;
}
Function call:
x = sum(3,4);
Pass by value:
void f(int x) { x = 10; }
int a = 5;
f(a);
// a stays 5
Pass by pointer (changes original):
void f(int *p) { *p = 10; }
int a = 5;
f(&a);
// a becomes 10
Prototype:
void process(int a[], int n);
Definition:
void process(int a[], int n) {
for(int i=0;i<n;i++) a[i] += 1;
}
Call:
process(a, n);
// array a is modified
Example:
double avg(int a[], int n) {
double s=0;
for(int i=0;i<n;i++) s+=a[i];
return s/n;
}
Find max:
int maxv(int a[], int n) {
int m=a[0];
for(int i=1;i<n;i++)
```

```
if(a[i]>m) m=a[i];
return m;
}
Count positives:
int countPos(int a[], int n){
int c=0;
for(int i=0;i<n;i++)
if(a[i]>0) c++;
return c;
}
Modify through pointer:
void inc(int *p){
*p = *p + 1;
}
Swap using pointers:
void swap(int *a, int *b){
int t=*a;
*a=*b;
*b=t;
}
Example:
void f(double *p, int *q){
*p = *p + 2.0;
*q = *q + 5;
}
double x=3.0;
int y=7;
f(&x,&y);
// x=5.0, y=12
Example:
void makeEven(int a[], int n){
for(int i=0;i<n;i++)
if(a[i]%2==1) a[i]++;
}
a={1,4,7} → makeEven → {2,4,8}
Files IO
Include library:
#include <fstream>
Open files:
ifstream fin("input.txt");
ofstream fout("output.txt");
Read values:
fin >> x;
while(fin >> a[n]) n++; // read entire file
into array
Read pairs:
while(fin >> x >> y) { ... }
Write values:
fout << x << endl;
Files behave like cin/cout but
read/write from/to a file.
Close files (optional):
fin.close();
fout.close();
Typical pattern:
n=0;
while(fin >> a[n]) n++;
for(int i=0;i<n;i++) fout << a[i] << endl;
Compute while reading:
while(fin >> x) fout << x*x << endl;
Read integers from a file, replace
negatives with 0, print results to
another file.
Solution outline:
n=0;
while(fin >> a[n]) n++;
for(i=0;i<n;i++)
```

```
if(a[i] < 0) a[i] = 0;
for(i=0;i<n;i++)
fout << a[i] << endl;
File contains pairs (x,y). For each pair,
compute x*x + y and write it out.
Solution outline:
while(fin >> x >> y)
fout << (x*x + y) << endl;
Read all numbers, compute sum and
average, print both.
Solution outline:
n=0;
while(fin >> a[n]) n++;
sum=0;
for(i=0;i<n;i++) sum += a[i];
avg = sum / n;
fout << sum << " " << avg;
Increment operators :
x++ : return x, then x = x+1
++x : x = x+1, then return x
int x=5, y;
y = x++; // y=5, x=6
y = ++x; // y=7, x=7
Switch
switch(x){
case 1: ... ; break;
case 2: ... ; break;
default: ... ;
}
Function prototype
Only if a function is defined after main,
declare prototype before main
int sum(int a, int b); // prototype
int sum(int a, int b){ // definition
return a+b;
}
Void function example :
void printArray(int a[], int n){
for(int i=0;i<n;i++) cout<<a[i]<<" ";
}
Local vs Global
int g = 10; // global
void f(){
int x = 5; // local
}
Compile-time
missing ;
undeclared variable
wrong type
Run-Time
divide by 0
out-of-bounds array access
uninitialized pointer
infinite loop
Logic errors
wrong condition
wrong formula
off-by-one loop
Pass by value
void f(int x){ x = 10; }
int a = 5;
f(a); // a still 5
Pass by reference (pointer)
void f(int *p){ *p = 10; }
int a = 5;
f(&a); // a becomes 10
```



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